

INPUT	PROCESS	OUTPUT
Name Contact Valkona Wireless headphones Monthly Instalment Desktop	Subtotal = desktop * (discount + laptop + Monthly Instalment) discount = Total * 0.3D total = (subtotal – discount) vat = total * 15% amount due = total + vat	Display results Name Contact Discount Total Vat

Pseudocode

This program will be used to calculate the costs of Valkona Headset and Desktop

Input (Provide the Selection for the Desktop)

Display “Provide the Selection for the desktop”

Select Laptop

Input (Provide the Selection for the Valkona wireless headset)

Display “Provide the Selection for the valkona wireless headset”

Select TV

Input (Provide the Selection for the Valkona wireless headset)

Display “Provide the Selection for the Monthly Instalment”

Select Monthly Instalment

Processing (Calculate Items bought)

total = ValkonaheadsetAmount + DesktopAmount

Processing (Calculate Items instalment)

instalment = total / terms

Processing (Calculating discount)

TotalLessThan16000D = 0.1 discount

TotalLessThan25000D = 0.2D discount

TotalLessthan30000D = 0.3D discount

Display “Customer name :", Name

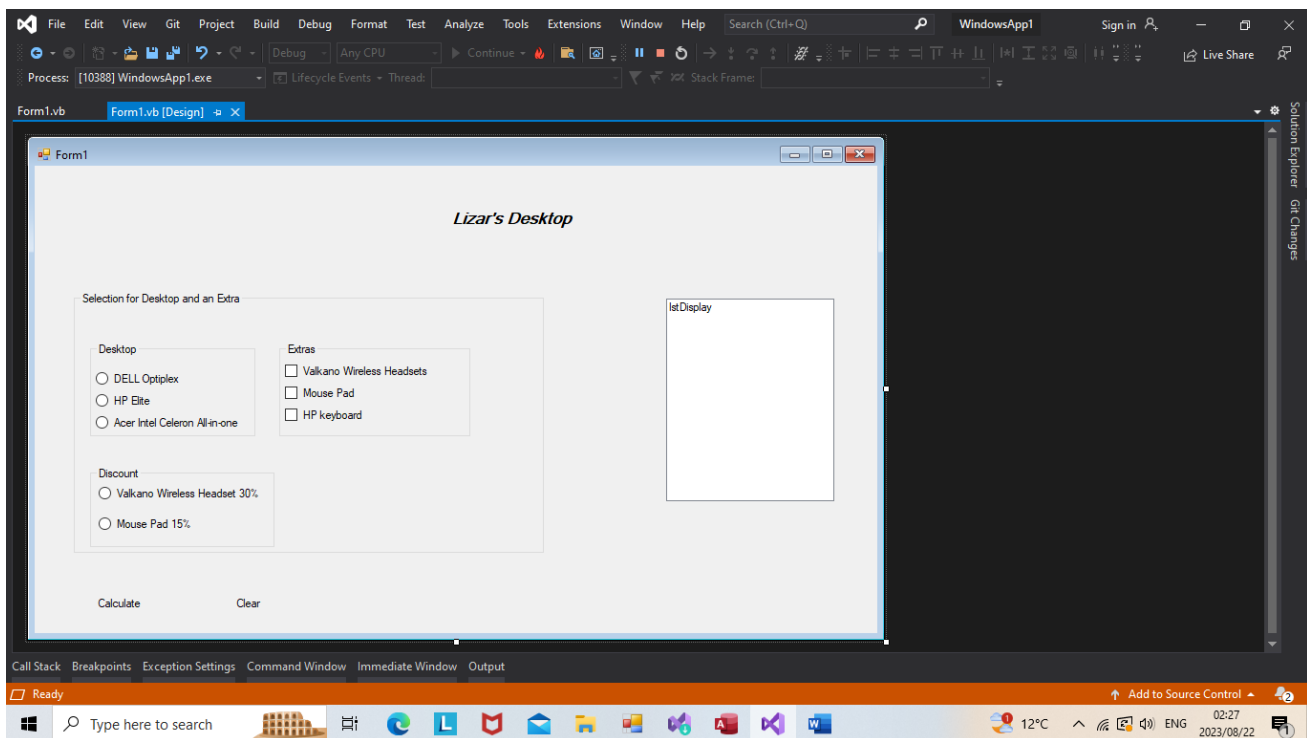
Display "Customer Contact:", Contact

Display "Total : " Total

Display " Monthly Installment :", Installation

Display "Discount :", Discount

3.1.2 GUI DESIGN



3.1.3 DEFENSIVE PROGRAMMING

Defensive programming is a form of defensive design intended to ensure the continuing function of a piece of software under unforeseen circumstances. Defensive programming practices are often used where high availability, safety or security is needed.

It improve software and source code, in terms of:

- General quality

- Making the source code comprehensible
- Making the software behave in a predictable manner

Overly defensive programming, however, may safeguard against errors that will never be encountered, thus incurring run-time and maintenance costs. There is also the risk that the code traps or prevents too many exceptions, potentially resulting in unnoticed, incorrect results.

Secure programming is the subset of defensive programming concerned with computer security. That is to say, security is the concern, not necessarily safety or availability. As with all kinds of defensive programming, avoiding bugs is a primary objective, however the motivation is not as much to reduce the likelihood of failure in normal operation, but to reduce the attack surface.

Offensive Programming is a category of defensive programming, with the added emphasis that certain errors should not be handled defensively. In this practice, only error from outside the program's control are to be handled (such as user input); the software itself, as well as data from within the program's line of defense, are to be trusted in this methodology.

- Encrypt & authenticate all important data transmitted over networks.
- Do not attempt to implement your own encryption scheme, use a proven one instead.
- All data is important until proven otherwise.
- All data is tainted until proven otherwise.
- All code is insecure until proven otherwise.

DATA VALIDATION

Data validation is the process of ensuring, at least as far as is possible, that the data given to a program by a user or from a file (essentially, the program's *input*) is of the correct type, and in the correct format. Although the programmer will obviously take every precaution to ensure the correct operation of the program, and will attempt to eliminate bugs that could cause a problem through a rigorous process of testing, they have no real control over mistakes made by the user during data entry. Nor can they guarantee that any data files used by the program will be free of errors and in the correct format.

There are however measures that can be taken to restrict the program's input to valid data. Such measures involve the application of *validation rules* to any data being input to the program. Input not meeting the program's requirements (i.e. input that does not obey the validation rules) can be dealt with in a pre-defined way, and will not cause the program to crash or produce spurious output. Data validation rules can also make an application more user friendly, since they enable the program to warn the user immediately when there is a problem rather than

simply allowing them to continue entering data until the program crashes or some other problem occurs.

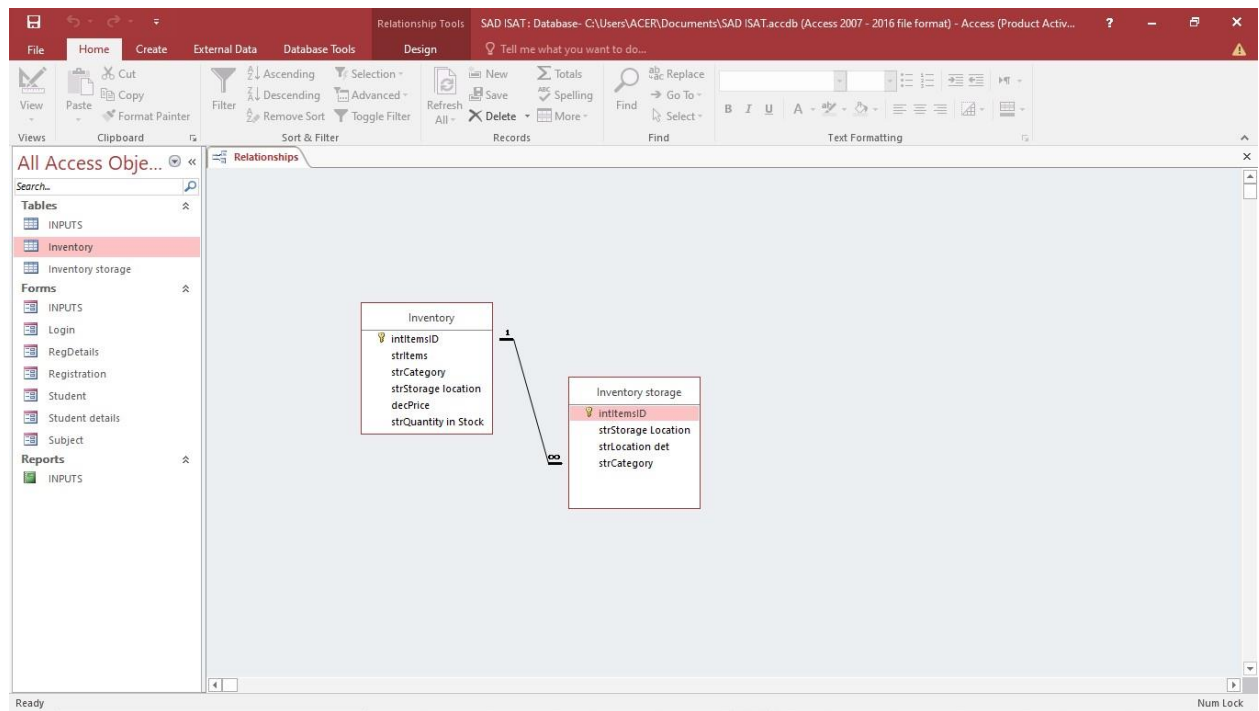
The validation rules used will depend on the application, and in particular on the type of data being input to the program. Database management systems like Oracle or Microsoft Access provide extensive facilities for creating data validation rules. For our VB.Net applications, we will need to work a little harder to ensure that the input is valid, because we will need to define and code our own validation rules.

In a software project of any size, the task of determining what *is* and what *isn't* valid data is usually undertaken long before any code is written, and is part of an initial requirements analysis phase. Even so, we can think in general terms about the kind of data validation required for a typical application, and how it might be implemented. Some common data validation requirements are outlined below.

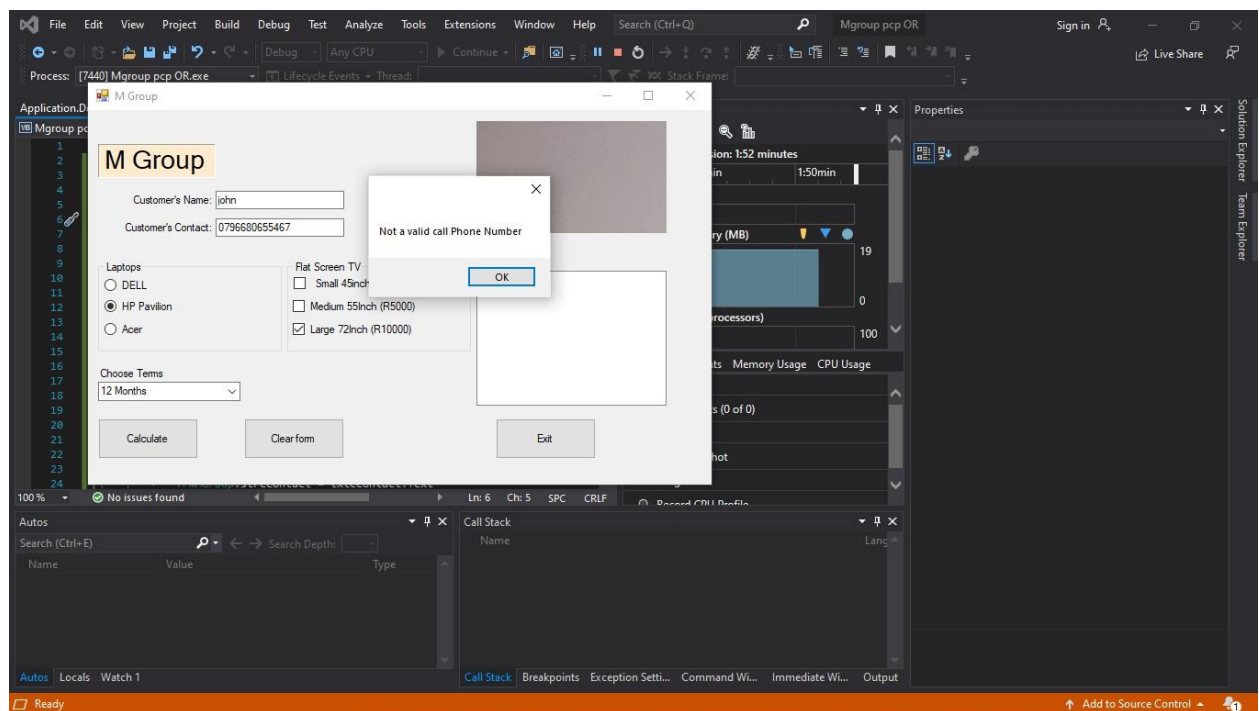
3.1.4

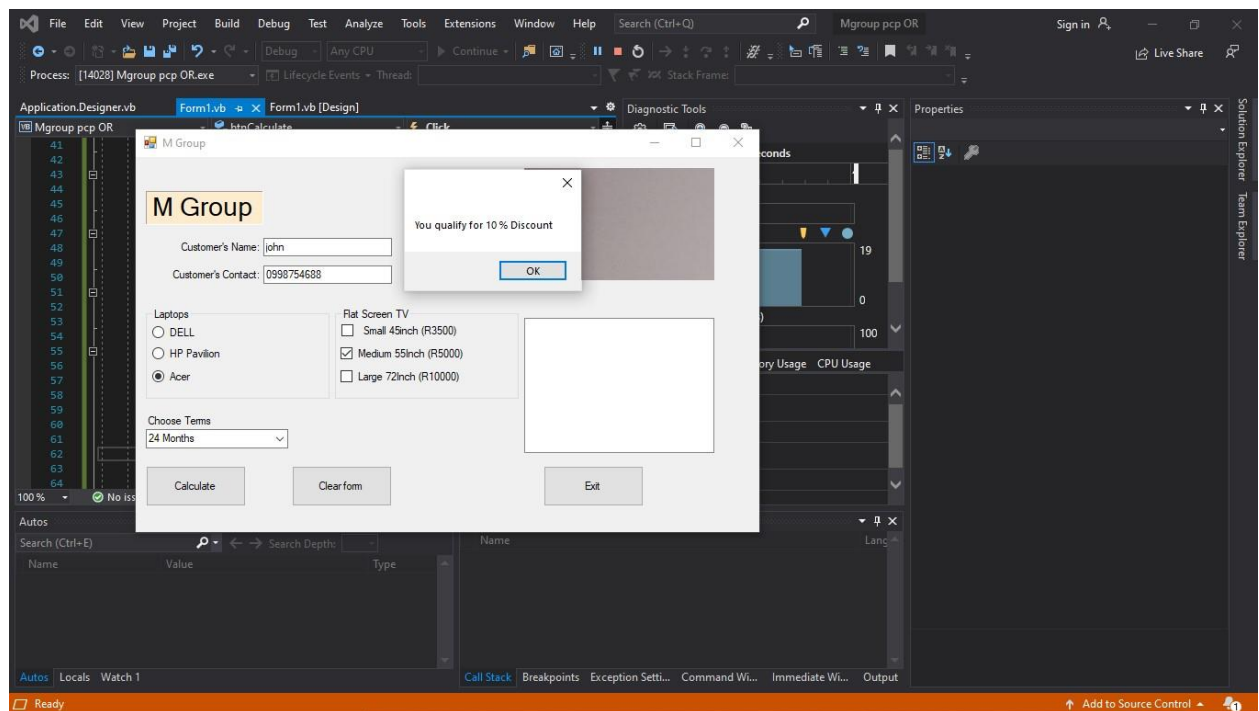
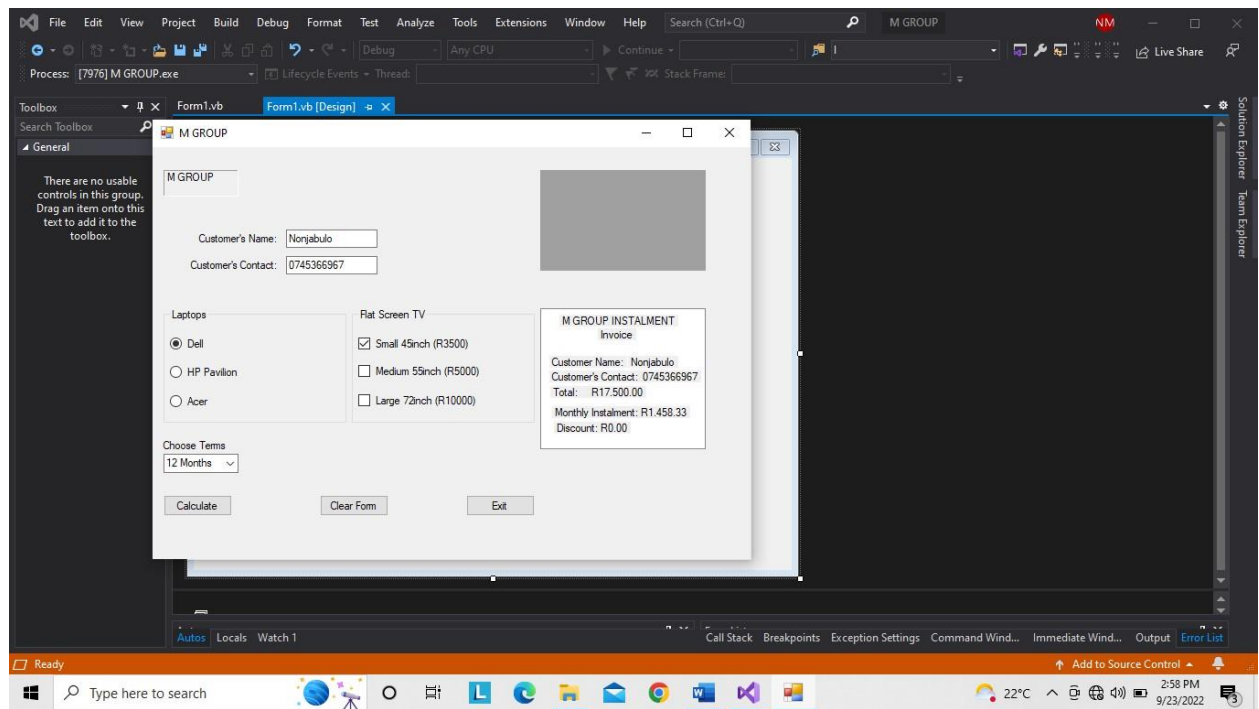
INPUT	PROCESS	OUTPUT
Valkona wireless headset Desktop	Subtotal = number of Desktop* (discount + Valkona headset) discount = 30% discount if buying both total = (subtotal – discount) vat = total * 15% amount due = total + vat	Display results Subtotal Discount Total Vat Amount due

3.1.5 EDR DESIGN



3.2.1





Public Class frmMGroup

Public Shared strName As String = ""

Public Shared strCContact As String = ""

Private Sub btnCalculate_Click(sender As Object, e As EventArgs) Handles btnCalculate.Click

Dim ValkonaheadsetPrice As Decimal = 0

Dim DesktopTotal As Decimal = 0

Dim Total As Decimal = 0

Dim decInstallation As Decimal = 0

Dim decDiscount As Decimal

Dim Term As Integer = 0

If IsNumeric(txtCName.Text) And (txtCName.TextLength > 3) Then

frmMGroup.strName = txtCName.Text

Else

MessageBox.Show(" Enter your Name")

txtCName.Focus()

End If

If IsNumeric(txtCContact.Text) And (txtCContact.TextLength = 10) Then

frmMGroup.strCContact = txtCContact.Text

Else

MessageBox.Show("Not a valid call Phone Number")

txtCContact.Focus()

End If

If chkSmall.Checked Then

DesktopTotal += 3500D

End If

If chkMedium.Checked Then

DesktopTotal += 5000D

End If

If chkLarge.Checked Then

DesktopTotal += 10000D

End If

If Total > 20000D Then

MessageBox.Show("You qualify for 10 % Discount")

decDiscount = Total * 0.1

Elseif Total > 35000D Then

MessageBox.Show("You qualify for 20 % Discount")

decDiscount = Total * 0.2

Elseif Total > 45000D Then

MessageBox.Show("You qualify for 30 % Discount")

decDiscount = Total * 0.3

Else

MessageBox.Show("You don't qualify for Discount")

decDiscount = 0

End If

'First item is 12 months

'Second item is 24 months

'Third item is 36 months

If cbTerms.SelectedIndex = 0 Then

Term = 12

Elseif cbTerms.SelectedIndex = 1 Then

Term = 24

Else

Term = 36

End If

Total = Valkona + DesktopTotal

decInstallation = Total / Term vat =

total * 15%

IstDisplay.Items.Add("M GROUP INSTALLMENT")

IstDisplay.Items.Add(vbTab & "Invoice") IstDisplay.Items.Add("")

IstDisplay.Items.Add(" Customer name :" & txtCName.Text)

IstDisplay.Items.Add("Customer Contact:" & txtCContact.Text)

IstDisplay.Items.Add("Total :" & Total.ToString("C2"))

IstDisplay.Items.Add(" Monthly Installment :" & decInstallation.ToString("C2"))

IstDisplay.Items.Add("") IstDisplay.Items.Add("Discount :" &

decDiscount.ToString("C2"))

End Sub

```
Private Sub btnClear_Click(sender As Object, e As EventArgs) Handles btnClear.Click
    rbDell.Checked = True      rbHP.Checked = False      rbPackard.Checked = False
    chkSmall.Checked = False   chkMedium.Checked = False   chkLarge.Checked =
    False      cbTerms.Items.Clear()      lstDisplay.Items.Clear()      txtCName.Text = ""
    txtCContact.Text = ""
```

End Sub

```
Private Sub btnExit_Click(sender As Object, e As EventArgs) Handles btnExit.Click
    Me.Close()
End Sub
```

End Class

3.3.2 TWO TABLES OF ERD

PCP ISAT SUBTASK #: Database- C:\Users\ACER\Documents\PCP ISAT SUBTASK #.accdb (Access 2007 - 2016 file format)...

File Home Create External Data Database Tools Fields Table Tell me what you want to do...

View Paste Copy Cut Copy Paste Format Painter

Views Clipboard

Sort & Filter

Filter Ascending Descending Advanced Remove Sort Toggle Filter

Records

Refresh Save Delete New Spelling More

Find Replace Go To Select

Calibri (Detail) 11

B I U Text Formatting

All Access Objects

Search...

Tables

strinventory

strinventory storage

strinventory storage

strCategory	strStorage Location	strLocation c	Click to Add
Candy	SE100A	Cooler	
Clothing	SE100B	Bins 1-10	
Miscellaneous	SE100C	Bins 1-2	
School supplies	SE100D	Bins 1-10	
Sports	SE100E	Bins 1-6	
*			

Record: 1 of 6

Datasheet View

PCP ISAT SUBTASK #: Database- C:\Users\ACER\Documents\PCP ISAT SUBTASK #.accdb (Access 2007 - 2016 file format)...

File Home Create External Data Database Tools Fields Table Tell me what you want to do...

View Paste Copy Cut Copy Paste Format Painter

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Find Replace Go To Select

Calibri (Detail) 11

B I U Text Formatting

All Access Objects

Search...

Tables

strinventory

strinventory storage

strinventory storage

strInventoryID	strItems	strCategory	strStorage location	decPrice	strQuantity in Stoc	Click to Add
1	Baseball	Sports	SE100D	\$0.00	50	
2	Soccerball	Sports	SE100D	\$7.99	10	
3	Chocolate bar	Candy	SE100A	\$0.89	250	
4	Lollipop	Candy	SE100A	\$0.50	500	
5	Ball Cap with logo	Clothing	SE100B	\$22.99	50	
6	Visor with Logo	Clothing	SE100B	\$11.25	50	
7	Pennant, Small	Sports	SE100D	\$6.00	100	
8	Pennant, Large	Sports	SE100D	\$12.00	50	
9	Notebook	School supplies	SE100C	\$4.75	250	
10	Pen with logo	School supplies	SE100C	\$2.25	500	
*				\$0.00	0	

Record: 1 of 12

Datasheet View

Num Lock

PCP ISAT SUBTASK #: Database- C:\Users\ACER\Documents\PCP ISAT SUBTASK #.accdb (Access 2007 - 2016 file format) ...

File Home Create External Data Database Tools Fields Table Tell me what you want to do...

View Paste Copy Format Painter Filter Sort & Filter Selection Advanced Refresh All Delete Records Find Replace Go To Select Text Formatting

All Access Objects

Tables

strinventory

strinventory storage

intItemsID	strItems	strCategory	strStorage location	decPrice	strQuantity in Stoc	Click to Add
4	Lollipop	Candy	SE100A	\$0.50	500	
11	Pencil with logo	School supplies	SE100C	\$0.75	1000	
3	Chocolate bar	Candy	SE100A	\$0.89	250	
10	Pen with logo	School supplies	SE100C	\$2.25	500	
17	Water bottle	Miscellaneous	SE100E	\$3.49	250	
9	Notebook	School supplies	SE100C	\$4.75	250	
7	Pennant, Small	Sports	SE100D	\$6.00	100	
1	Baseball	Sports	SE100D	\$7.99	50	
6	Visor with Logo	Clothing	SE100B	\$11.25	50	
8	Pennant, Large	Sports	SE100D	\$12.00	50	
2	Soccerball	Sports	SE100D	\$15.99	10	
12	T-Shirt	Clothing	SE100B	\$17.50	100	
13	T-Shirt, M	Clothing	SE100B	\$18.50	100	
14	T-Shirt, L	Clothing	SE100B	\$19.50	100	
15	T-Shirt, XL	Clothing	SE100B	\$20.50	100	
16	T-Shirt, XXL	Clothing	SE100B	\$21.50	100	
5	Ball Cap with logo	Clothing	SE100B	\$22.99	50	
*	0			\$0.00	0	

Record: 14 of 18 of 18 No Filter Search

Datasheet View

PCP ISAT SUBTASK #: Database- C:\Users\ACER\Documents\PCP ISAT SUBTASK #.accdb (Access 2007 - 2016 file format) ...

File Home Create External Data Database Tools Fields Table Tell me what you want to do...

View Paste Copy Format Painter Filter Sort & Filter Selection Advanced Refresh All Delete Records Find Replace Go To Select Text Formatting

All Access Objects

Tables

strinventory

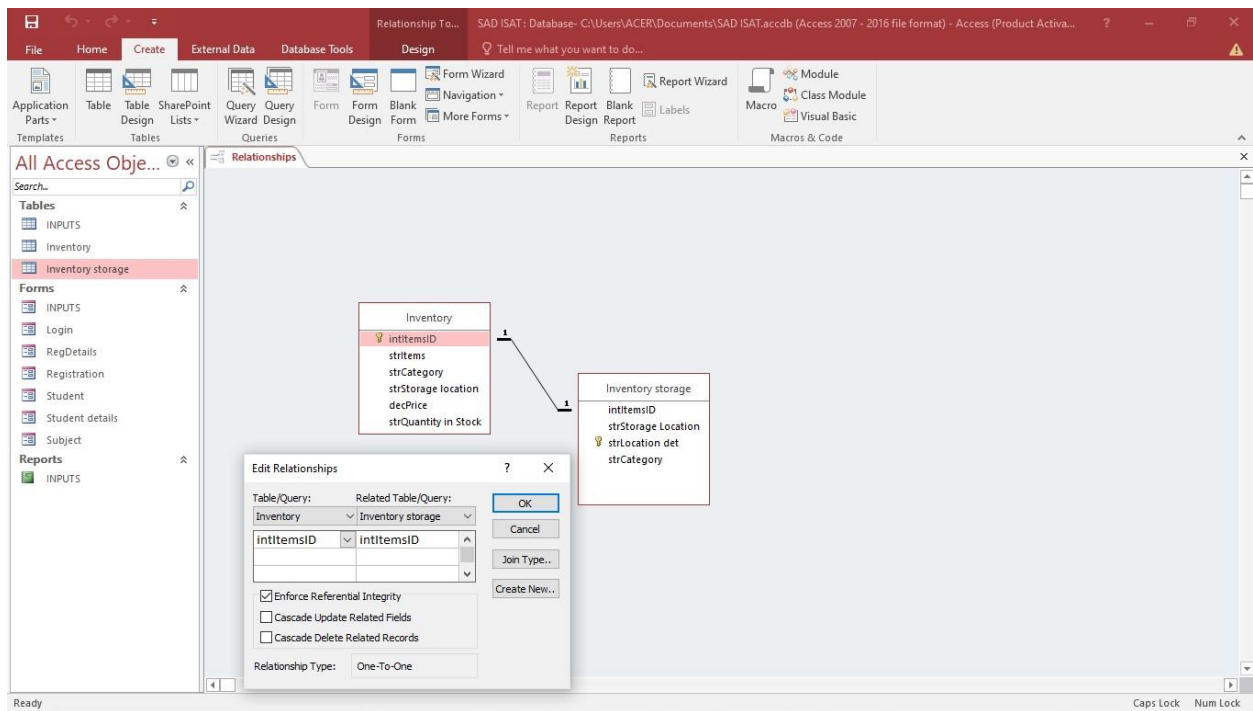
strinventory storage

strCategory	strStorage Location	strLocation	Click to Add
Candy	SE100A	Cooler	
Clothing	SE100B	Bins 1-10	
Miscellaneous	SE100C	Bins 1-2	
School supplies	SE100D	Bins 1-10	
Sports	SE100E	Bins 1-6	
*			

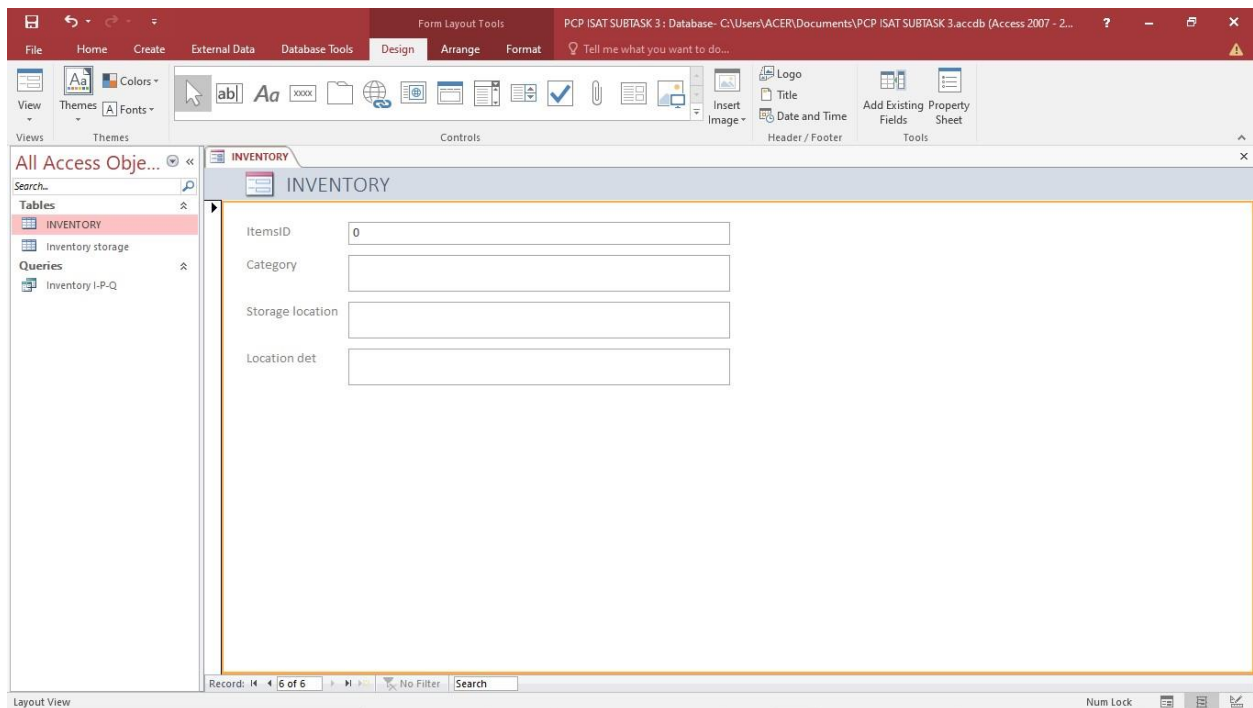
Record: 14 of 6 of 6 No Filter Search

Datasheet View

3.3.4 REFERENTIAL INTEGRITY



3.3.5 FIRST FORM



PCP ISAT SUBTASK 3 : Database- C:\Users\ACER\Documents\PCP ISAT SUBTASK 3.accdb (Access 2007 - 2...

File Home Create External Data Database Tools Design Arrange Format Tell me what you want to do...

Form

Select All B I U A - Font Formatting \$ % +.0 -0.0 Background Image Alternate Row Color Quick Change Conditional Styles Shape Formatting Shape Effects Shape Fill Shape Outline Shape Effects

Selection

All Access Objects

Search...

Tables

INVENTORY

Inventory storage

Queries

Inventory I-P-Q

Inventory storage

ItemsID 0

Items

Category

Storage Location

Price \$0.00

Quantity in stock 0

Record: 11 of 11 No Filter Search

Layout View Num Lock

PCP ISAT SUBTASK 3 : Database- C:\Users\ACER\Documents\PCP ISAT SUBTASK 3.accdb (Access 2007 - 2...

File Home Create External Data Database Tools Design Arrange Format Tell me what you want to do...

View Themes Colors Themes

Views Themes

All Access Objects

Search...

Tables

INVENTORY

Inventory storage

Queries

Inventory I-P-Q

Inventory storage

ItemsID 0

Items

Category

Storage Location

Price \$0.00

Quantity in stock 0

ADD

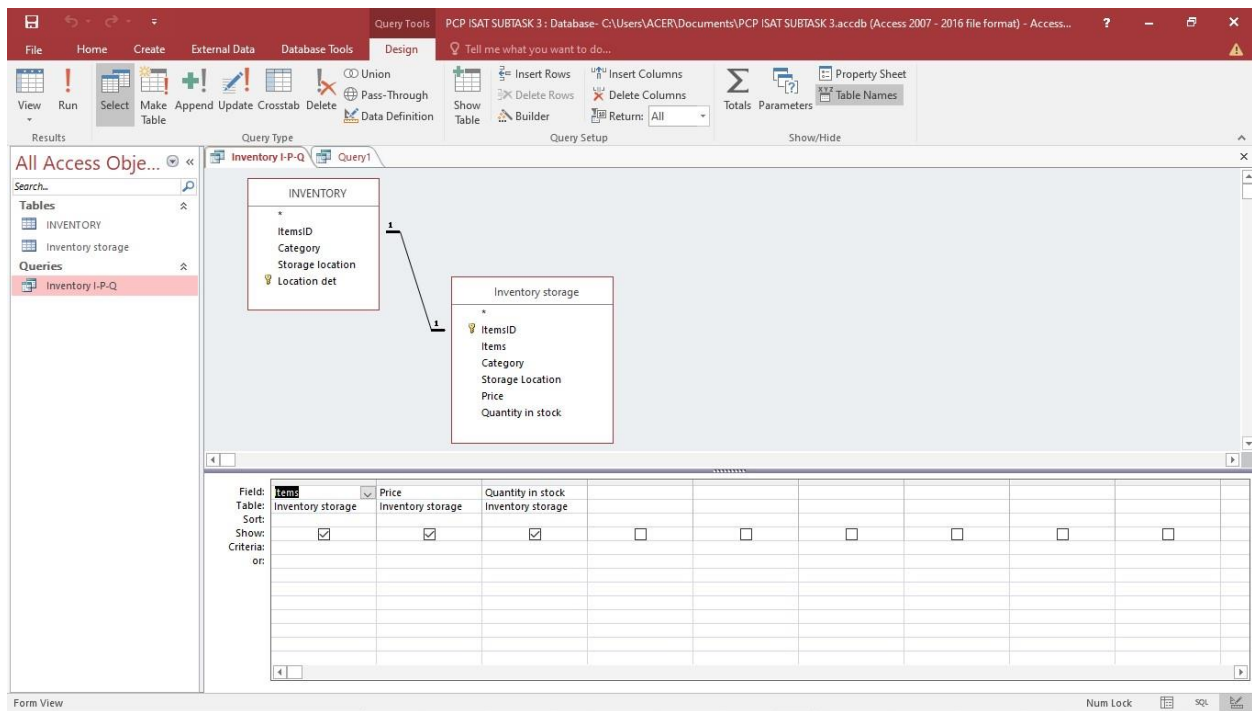
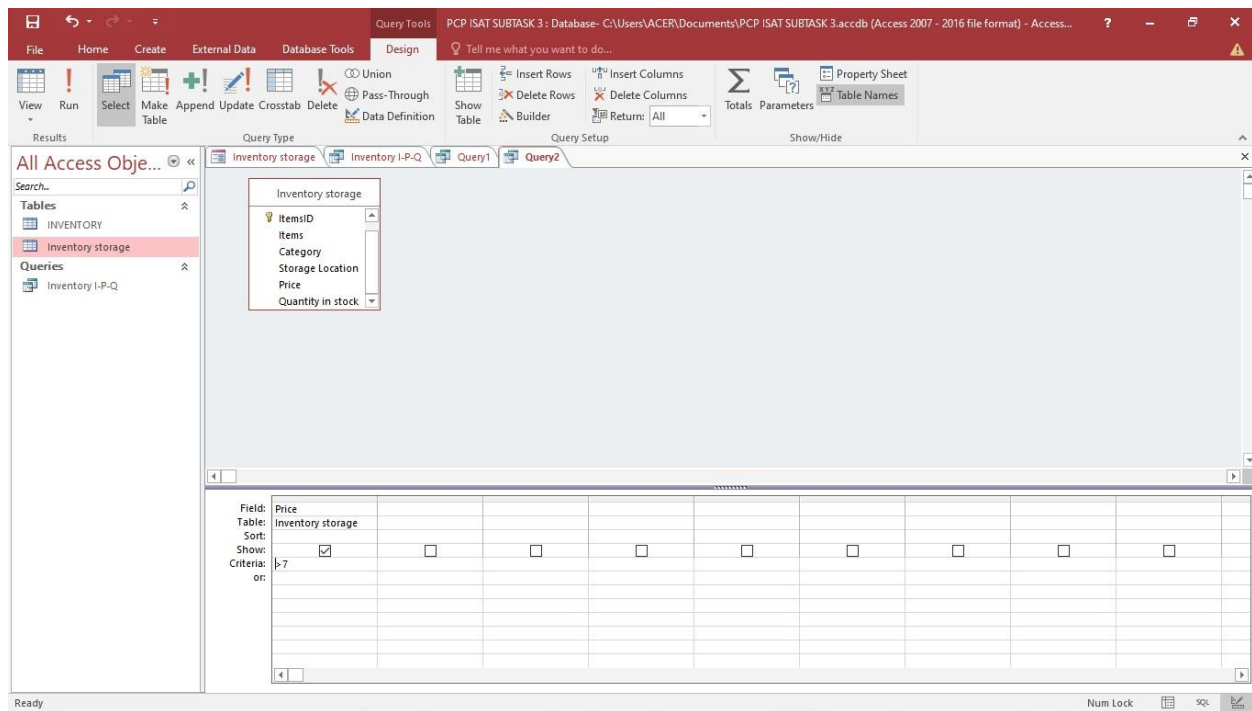
Exit

Record: 11 of 11 No Filter Search

Layout View Num Lock

3.3.6 Queries

Ready



PCP ISAT SUBTASK 3: Database- C:\Users\ACER\Documents\PCP ISAT SUBTASK 3.accdb (Access 2007 - 2016 file format) - Access...

File Home Create External Data Database Tools Design Tell me what you want to do...

View Run Select Make Table Append Update Crosstab Delete Union Pass-Through Data Definition Show Table Builder Return: All Totals Parameters Property Sheet Table Names

Query Type Inventory I-P-Q Query1

All Access Objects

Search...

Tables

- INVENTORY
- Inventory storage

Queries

- Inventory I-P-Q

INVENTORY

- ItemsID
- Category
- Storage location
- Location det

Inventory storage

- ItemsID
- Items
- Category
- Storage Location
- Price
- Quantity in stock

Field: Items Table: Inventory storage Price Inventory storage Quantity in stock Inventory storage

Show: ☒ ☒ ☒

Criteria: ☐ ☐ ☐

or: ☐ ☐ ☐

Form View

Num Lock SQL

PCP ISAT SUBTASK 3: Database- C:\Users\ACER\Documents\PCP ISAT SUBTASK 3.accdb (Access 2007 - 2016 file format) - Access (Product Activation Failed)

File Home Create External Data Database Tools Tell me what you want to do...

View Paste Copy Cut Filter Sort & Filter Records Find Text Formatting

Inventory I-P-Q Query1

All Access Objects

Search...

Tables

- INVENTORY
- Inventory storage

Queries

- Inventory I-P-Q

Items	Price	Quantity in stock
Baseball	\$7.99	50
Soccerball	\$15.99	10
Ball cap with Ic	\$22.99	50
*		

Record: 1 of 3 No Filter Search

Datasheet View

Num Lock SQL

